Project Design Phase
Solution Architecture

Date	15 February 2025
Team ID	LTVIP2026TMIDS54144
Project Name	ShopSmart – Digital Grocery Web Application
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

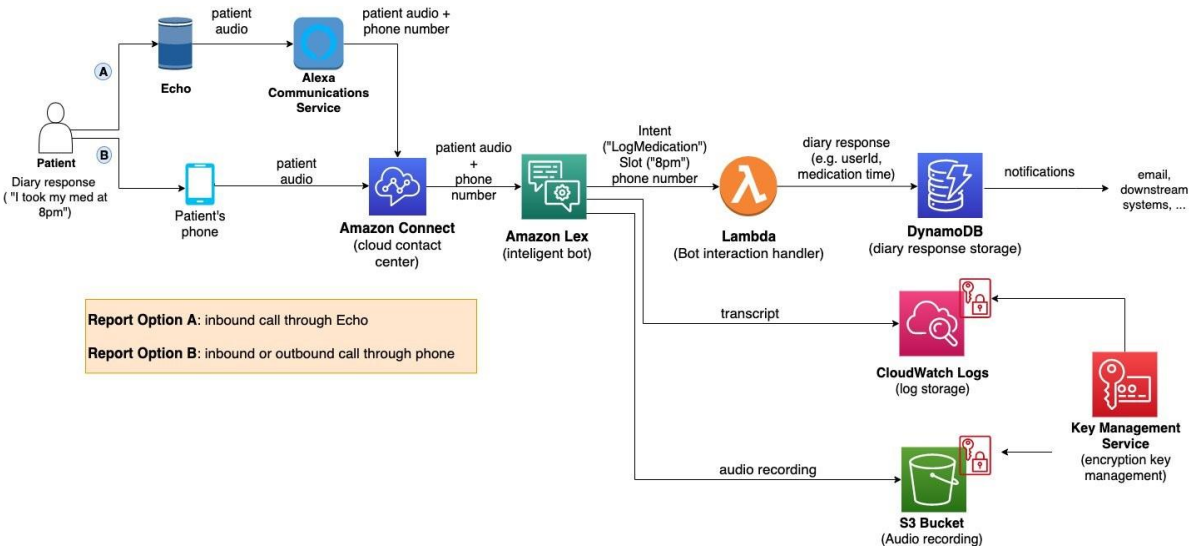

Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

A. ARCHITECTURE OVERVIEW

Use 3-tier architecture:

1. Presentation Layer:
- Angular frontend
- Responsive UI
- Bootstrap styling

2. Application Layer:
- Node.js + Express
- REST APIs
- Authentication middleware
- Business logic modules

3. Data Layer:
- MongoDB database
- User collection
- Product collection
- Orders collection

B. EXTERNAL INTERFACES

- Stripe Payment Gateway
- MongoDB Atlas (Cloud database)
- Email service for confirmations

C. INFRASTRUCTURE

Local:
- Development server
- Local MongoDB

Cloud:
- MongoDB Atlas
- Backend hosted on Render/Railway
- Frontend hosted on Netlify/Vercel

D. DATA FLOW EXPLANATION

1. User sends request via Angular UI
2. API request goes to Express server
3. Server validates JWT
4. Server interacts with MongoDB
5. Payment processed via Stripe
6. Order confirmation returned to frontend

E. CHARACTERISTICS

Security:
- JWT authentication
- bcrypt password hashing
- Environment variables
- HTTPS ready

Scalability:
- Modular backend
- Cloud-ready deployment

Availability:
- Cloud database uptime
- Independent frontend & backend hosting

Performance:
- Indexed MongoDB queries
- Optimized API routing